Hopf algebras and tensor categories

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Plan of the talk.

I. Hopf algebras.

II. Tensor categories.

III. Classification of Hopf algebras: state of the art.

I. Hopf algebras. $\Bbbk$ algebraically closed field

A algebra: product $\mu : A \otimes A \to A$, unit $u : \Bbbk \to A$

Associative:

$$\begin{tikzcd}
A \otimes A \otimes A \arrow[r, "\mathrm{id} \otimes \mu"] \arrow[d, "\mu \otimes \mathrm{id}"'] & A \otimes A \arrow[d, "\mu"] \\
A \otimes A \arrow[r, "\mu"] & A
\end{tikzcd}$$

Unitary:

$$\begin{tikzcd}
\Bbbk \otimes A \arrow[r, "u \otimes \mathrm{id}"] \arrow[dr, "\sim"'] & A \otimes A \arrow[d, "\mu"] & A \otimes \Bbbk \arrow[l, "\mathrm{id} \otimes u"'] \arrow[dl, "\sim"] \\
& A &
\end{tikzcd}$$

C coalgebra: coproduct $\Delta : C \to C \otimes C$, counit $\varepsilon : C \to \Bbbk$

Co-associative:

$$\begin{tikzcd}
C \otimes C \otimes C & C \otimes C \arrow[l, "\mathrm{id} \otimes \Delta"'] \\
C \otimes C \arrow[u, "\Delta \otimes \mathrm{id}"] & C \arrow[l, "\Delta"'] \arrow[u, "\Delta"']
\end{tikzcd}$$

Co-unitary:

$$\begin{tikzcd}
\Bbbk \otimes C \arrow[dr, "\sim"'] & C \otimes C \arrow[l, "\varepsilon \otimes \mathrm{id}"'] \arrow[r, "\mathrm{id} \otimes \varepsilon"] & C \otimes \Bbbk \arrow[dl, "\sim"] \\
& C \arrow[u, "\Delta"] &
\end{tikzcd}$$

Hopf algebra: $(H, \mu, u, \Delta, \varepsilon)$

- (H, μ, u) algebra
- (H, Δ, ε) coalgebra
- Δ, ε algebra maps
- There exists $\mathcal{S} : H \to H$ (the antipode) such that

$$\begin{tikzcd}
H \arrow[r, "\Delta"] \arrow[dr, "\varepsilon"'] & H \otimes H \arrow[r, "\mathcal{S}\otimes\mathrm{id}", "\mathrm{id}\otimes\mathcal{S}"'] & H \otimes H \arrow[r, "\mu"] & H \\
& \Bbbk \arrow[ur, "u"'] & &
\end{tikzcd}$$

Example:

- Γ finite group
- $H = \mathcal{O}(\Gamma) =$ algebra of functions $\Gamma \to \Bbbk$
- $\Delta : H \to H \otimes H \simeq \mathcal{O}(\Gamma \times \Gamma)$, $\Delta(f)(x, y) = f(x.y)$.
- $\varepsilon : H \to \Bbbk$, $\varepsilon(f) = f(e)$.
- $\mathcal{S} : H \to H$, $\mathcal{S}(f)(x) = f(x^{-1})$.

Remark: $(H, \mu, u, \Delta, \varepsilon)$ finite-dimensional Hopf algebra
$\Longrightarrow$ $(H^*, \Delta^t, \varepsilon^t, \mu^t, u^t)$ Hopf algebra

Example: $H = \mathcal{O}(\Gamma)$; for $x \in \Gamma$, $E_x \in H^*$, $E_x(f) = f(x)$. Then

$$E_x E_y = E_{xy}, \qquad \mathcal{S}(E_x) = E_{x^{-1}}.$$

Hence $H^* = \Bbbk\Gamma$, group algebra of Γ.

Remark: $(H, \mu, u, \Delta, \varepsilon)$ Hopf algebra with $\dim H = \infty$,
H^* NOT a Hopf algebra,
but contains a largest Hopf algebra with operations transpose to those of H.

Examples:

- Γ affine algebraic group

- $H = \mathcal{O}(\Gamma) =$ algebra of regular (polynomial) functions $\Gamma \to \Bbbk$ is a Hopf algebra with analogous operations.

- $H^* \supset \Bbbk\Gamma$
- $H^* \supset \mathcal{U} :=$ algebra of distributions with support at e; this is a Hopf algebra

- If $\operatorname{char} \Bbbk = 0$, then $\mathcal{U} \simeq U(\mathfrak{g})$, $\mathfrak{g} =$ Lie algebra of Γ

- If $\mathfrak{g}$ is any Lie algebra, then the enveloping algebra $U(\mathfrak{g})$ is a Hopf algebra with $\Delta(x) = x \otimes 1 + 1 \otimes x$, $x \in \mathfrak{g}$.

Short history:

• Since the dictionary *Lie groups* $\leftrightsquigarrow$ *Lie algebras* fails when char > 0, Dieudonné studied in the early 50's the hyperalgebra $\mathcal{U}$. Pierre Cartier introduced the abstract notion of hyperalgebra (cocommutative Hopf algebra) in 1955.

• Armand Borel considered algebras with a coproduct in 1952, extending previous work of Hopf. He coined the expression *Hopf algebra*.

• George I. Kac introduced an analogous notion in the context of von Neumann algebras.

• The first appearance (that I am aware of) of the definition as it is known today is in a paper by Kostant (1965).

First invariants of a Hopf algebra H:
$G(H) = \{x \in H - 0 : \Delta(x) = x \otimes x\}$, group of grouplikes.

$\mathrm{Prim}(H) = \{x \in H : \Delta(x) = x \otimes 1 + 1 \otimes x\}$, Lie algebra of primitive elements.

$\tau : V \otimes W \to W \otimes W$, $\tau(v \otimes w) = w \otimes v$ the *flip*.
H is commutative if $\mu\tau = \mu$. H is cocommutative if $\tau\Delta = \Delta$.

Group algebras, enveloping algebras, hyperalgebras are cocommutative.

Theorem. (Cartier-Kostant, early 60's). $\mathrm{char}\,\Bbbk = 0$.
Any cocommutative Hopf algebra is of the form $U(\mathfrak{g})\#\Bbbk\Gamma$.

$H = \Bbbk[X]$, $\Delta(X) = X \otimes 1 + 1 \otimes X$. Then

$$\Delta(X^n) = \sum_{0 \leq j \leq n} \binom{n}{j} X^j \otimes X^{n-j}.$$

If $\operatorname{char} \Bbbk = p > 0$, then $\Delta(X^p) = X^p \otimes 1 + 1 \otimes X^p$.

Thus $\Bbbk[X]/\langle X^p \rangle$, $\Delta(X) = X \otimes 1 + 1 \otimes X$ is a Hopf algebra, commutative and cocommutative, dim p.

(Kulish, Reshetikhin and Sklyanin, 1981). Quantum $SL(2)$: if $q \in \Bbbk$, $q \neq 0, \pm 1$, set

$$\begin{aligned} U_q(\mathfrak{sl}(2)) = \Bbbk\langle E, F, K, K^{-1} | KK^{-1} &= 1 = K^{-1}K \\ KE &= q^2 EK, \\ KF &= q^{-2} FK, \\ EF - FE &= \frac{K - K^{-1}}{q - q^{-1}} \rangle \end{aligned}$$

$$\begin{aligned} \Delta(K) &= K \otimes K, \\ \Delta(E) &= E \otimes 1 + K \otimes E, \\ \Delta(F) &= F \otimes K^{-1} + 1 \otimes F. \end{aligned}$$

These Hopf algebras, neither commutative nor cocommutative, are analogues of the enveloping algebra of $\mathfrak{sl}(2)$.

(Lusztig, 1989). If q is a root of 1 of order N odd, then

$$\begin{aligned} \mathfrak{u}_q(\mathfrak{sl}(2)) = \Bbbk\langle E, F, K, K^{-1} | &\text{same relations plus} \\ K^N = 1, \quad E^N = F^N &= 0\rangle. \end{aligned}$$

These Hopf algebras, neither commutative nor cocommutative, are analogues of the Frobenius kernel of $\mathfrak{sl}(2)$.

There are dual Hopf algebras, analogues of the algebra of regular functions of $SL(2)$.

$$\mathcal{O}_q(SL(2)) = \Bbbk\langle \begin{pmatrix} a & b \\ c & d \end{pmatrix} | ab = qba, \quad ac = qca, \quad bc = cb,$$
$$bd = qdb, \quad cd = qdc, \quad ad - da = (q - q^{-1})bc,$$
$$ad - qbc = 1\rangle.$$
$$\Delta \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \otimes \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

(Manin). If q is a root of 1 of order N odd, then

$$\mathfrak{o}_q(\mathfrak{sl}(2)) = \Bbbk\langle \begin{pmatrix} a & b \\ c & d \end{pmatrix} | \text{same relations plus}$$
$$a^N = 1 = d^N, \quad b^N = c^N = 0\rangle.$$

In 1983, Drinfeld and Jimbo introduced quantized enveloping algebras $U_q(\mathfrak{g})$, for q as above and $\mathfrak{g}$ any simple Lie algebra.

- Quantum function algebras $\mathcal{O}_q(G)$: Faddeev-Reshetikhin and Takhtajan (for $SL(N)$) and Lusztig (any simple G).

- Finite-dimensional versions when q is a root of 1.

II. Tensor categories.

Monoidal categories (categorical versions of groups).

A *monoidal category* is a category $\mathcal{C}$ provided with

- A bifuntor $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$, called *tensor*;
- an object $\mathbf{1} \in \mathcal{C}$, called *unit*;
- an *associativity* constraint, i. e. a natural isomorphism

$$a_{X,Y,Z} : (X \otimes Y) \otimes Z \to (X \otimes Y) \otimes Z;$$

- left and right *unit* constraints, i. e. natural isomorphisms

$$l_X : \mathbf{1} \otimes X \simeq X, \qquad r_X : X \otimes \mathbf{1} \simeq X.$$

$(\mathcal{C}, \otimes, a, \mathbf{1}, l, r)$ should satisfy the pentagon and triangle axioms, i. e. the commutativity of (1), (2), for any $X, Y, Z, W \in \mathrm{Obj}\,\mathcal{C}$:

$$\begin{tikzcd}
((X\otimes Y)\otimes Z)\otimes W \arrow[r, "a_{X\otimes Y,Z,W}"] \arrow[d, "a_{X,Y,Z}\otimes \mathrm{id}_W"] & (X\otimes Y)\otimes(Z\otimes W) \arrow[r, "a_{X,Y,Z\otimes W}"] & X\otimes(Y\otimes(Z\otimes W)) \\
(X\otimes(Y\otimes Z))\otimes W \arrow[rr, "a_{X,Y\otimes Z,W}"] & & X\otimes((Y\otimes Z)\otimes W) \arrow[u, "\mathrm{id}_X\otimes a_{Y,Z,W}"']
\end{tikzcd} \tag{1}$$

$$\begin{tikzcd}
(X\otimes\mathbf{1})\otimes Y \arrow[rr, "r_X\otimes \mathrm{id}_Y"] \arrow[dr, "a_{X,\mathbf{1},Y}"] & & X\otimes Y \\
& X\otimes(\mathbf{1}\otimes Y) \arrow[ur, "\mathrm{id}_X\otimes l_Y"'] &
\end{tikzcd} \tag{2}$$

Let $\mathcal{C}$ be a monoidal category.

A *right dual* of $V \in \operatorname{Obj}\mathcal{C}$ is a collection (V^*, e_V, b_V), where

- $V^* \in \operatorname{Obj}\mathcal{C}$,
- $e_V : V^* \otimes V \to \mathbf{1}$ is a morphism called *evaluation*,
- $b_V : \mathbf{1} \to V \otimes V^*$ is a morphism called *coevaluation*, such that

$$\begin{tikzcd} V \arrow[r, "l_V^{-1}"] \arrow[rrrrr, bend right, "\mathrm{id}_V"'] & \mathbf{1}\otimes V \arrow[r, "b_V \otimes \mathrm{id}_V"] & (V\otimes V^*)\otimes V \arrow[r, "a_{V,V^*,V}"] & V\otimes(V^*\otimes V) \arrow[r, "\mathrm{id}_V \otimes e_V"] & V\otimes\mathbf{1} \arrow[r, "r_V"] & V \end{tikzcd},$$

$$\begin{tikzcd} V^* \arrow[r, "r_{V^*}^{-1}"] \arrow[rrrrr, bend right, "\mathrm{id}_V^*"'] & V^*\otimes\mathbf{1} \arrow[r, "\mathrm{id}_{V^*} \otimes b_V"] & V^*\otimes(V\otimes V^*) \arrow[r, "a_{V^*,V,V^*}^{-1}"] & (V^*\otimes V)\otimes V^* \arrow[r, "e_V \otimes \mathrm{id}_{V^*}"] & \mathbf{1}\otimes V^* \arrow[r, "l_{V^*}"] & V^* \end{tikzcd}.$$

A *left dual* of $V \in \operatorname{Obj}\mathcal{C}$ is a collection $({}^*V, e'_V, b'_V)$, where

- ${}^*V \in \operatorname{Obj}\mathcal{C}$,
- $e'_V : {}^*V \otimes V \to \mathbf{1}$, $b'_V : \mathbf{1} \to V \otimes {}^*V$ are morphisms such that

$$\begin{tikzcd}
V \arrow[r, "r_V^{-1}"] \arrow[rrrrr, bend right, "\mathrm{id}_V"'] & V \otimes \mathbf{1} \arrow[r, "\mathrm{id}_V \otimes b'_V"] & V \otimes ({}^*V \otimes V) \arrow[r, "a^{-1}_{V,{}^*V,V}"] & (V \otimes {}^*V) \otimes V) \arrow[r, "e'_V \otimes \mathrm{id}_V"] & \mathbf{1} \otimes V \arrow[r, "l_V"] & V\,,
\end{tikzcd}$$

$$\begin{tikzcd}
{}^*V \arrow[r, "l_{{}^*V}^{-1}"] \arrow[rrrrr, bend right, "\mathrm{id}_{{}^*V}"'] & \mathbf{1} \otimes {}^*V \arrow[r, "b'_V \otimes \mathrm{id}_{{}^*V}"] & ({}^*V \otimes V) \otimes {}^*V \arrow[r, "a_{{}^*V,V,V^*}"] & {}^*V \otimes (V \otimes {}^*V) \arrow[r, "\mathrm{id}_{{}^*V} \otimes e'_V"] & {}^*V \otimes \mathbf{1} \arrow[r, "r_{{}^*V}"] & {}^*V.
\end{tikzcd}$$

A monoidal category is *rigid* if every object admits a right and a left dual.

Examples:

- $\mathcal{C}$ discrete category (only arrows are the identities)

monoidal $\leftrightsquigarrow$ monoid rigid monoidal $\leftrightsquigarrow$ group

- $\mathsf{Vec}_{\Bbbk}$ = category of vector spaces over $\Bbbk$, $\otimes = \otimes_{\Bbbk}$

$V \in \mathsf{Vec}_{\Bbbk}$ has duals ($V^* = \mathsf{Hom}(V, \Bbbk) = {}^*V$) $\iff$ $\dim V < \infty \rightsquigarrow$ $\mathsf{vec}_{\Bbbk}$ = category of fin. dim. vector spaces is rigid

- R a $\Bbbk$-algebra, Bimod_R = category of R-bimodules, $\otimes = \otimes_R$

- G a group, Rep_G, $\otimes = \otimes_{\Bbbk}$; rep_G = fin. dim. reps. is rigid

$V, W \in \mathsf{Rep}_G$, $v \in V, w \in W, g \in G$: $g \cdot (v \otimes w) = g \cdot v \otimes g \cdot w$.

- $\mathfrak{g}$ a Lie algebra, $\mathsf{Rep}_{\mathfrak{g}}$, $\otimes = \otimes_{\Bbbk}$; $\mathsf{rep}_{\mathfrak{g}}$ = fin. dim. reps. is rigid
$V, W \in \mathsf{Rep}_{\mathfrak{g}}$, $v \in V, w \in W, X \in \mathfrak{g}$: $X \cdot (v \otimes w) = X \cdot v \otimes w + v \otimes X \cdot w$.

- H a Hopf algebra with bijective antipode $\mathcal{S}$, Rep_H, $\otimes = \otimes_{\Bbbk}$; rep_H = fin. dim. reps. is rigid
- $V, W \in \mathsf{Rep}_H$, $v \in V, w \in W, X \in H$: set $\Delta(X) = \sum_i X_i \otimes X^i$, then

$$X \cdot (v \otimes w) = \sum_i X_i \cdot v \otimes X^i \cdot w.$$

- $\mathbf{1} := \Bbbk \in \mathsf{Rep}_H$: if $X \in H$, then $X \cdot 1 = \varepsilon(X) 1$.

- $V \in \mathsf{rep}_H \rightsquigarrow V^* = \mathsf{Hom}(V, \Bbbk) = {}^*V$ as v. sp. but with **different** actions: $v \in V, X \in H, \alpha \in \mathbf{V}^* = \mathsf{Hom}(V, \Bbbk), \beta \in {}^*\mathbf{V} = \mathsf{Hom}(V, \Bbbk)$

$$\langle X \cdot \alpha, v \rangle = \langle \alpha, \mathcal{S}(X) \cdot v \rangle, \qquad \langle X \cdot \beta, v \rangle = \langle \beta, \mathcal{S}^{-1}(X) \cdot v \rangle.$$

Tensor categories

A *tensor category* is a rigid monoidal category $\mathcal{C}$ such that

- $\mathcal{C}$ is abelian (good kernels and cokernels);
- $\mathcal{C}$ is $\Bbbk$-linear ($\mathrm{Hom}(V, W)$ is a $\Bbbk$-v. sp., composition is bilinear);
- the tensor product $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ is $\Bbbk$-bilinear;
- the unit $\mathbf{1} \in \mathcal{C}$ is simple and $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, \mathbf{1}) \simeq \Bbbk$;

Example: H a Hopf algebra with bijective antipode $\mathcal{S}$

$\rightsquigarrow$ rep_H is a tensor category

A *finite tensor category* is a finite category $\mathcal{C}$ such that

- there is a finite number of iso. classes of simple objects;
- $\dim \mathrm{Hom}(V, W) < \infty$ for all $V, W \in \mathrm{Obj}\,\mathcal{C}$;
- every object has a projective cover;
- the length of the Jordan-Hölder series of every object is finite.

Example: H a **finite dimensional** Hopf algebra

$\rightsquigarrow$ rep_H is a finite tensor category

Logarithmic CFT $\rightsquigarrow$ finite tensor categories

A *fusion category* is a **semisimple** finite tensor category $\mathcal{C}$. That is, any object is a direct sum of simple objects.

Rational CFT $\rightsquigarrow$ fusion categories

Example: H a semisimple Hopf algebra

$\rightsquigarrow$ rep_H is a fusion category

Let $\mathcal{C}$ be a tensor category (finite, fusion).

Assume that there exists a *fiber functor* $\mathcal{C} \to \mathrm{Vec}_{\Bbbk}$. Then $\exists$H Hopf alg.

$$\mathcal{C} \simeq \mathrm{rep}_H$$

as tensor categories (fin. dim., semisimple).

Note: H a Hopf algebra, $F \in (H \otimes H)^{\times}$ satisfying certain conditions. Then $H_F := (H, \Delta_F = F \Delta F^{-1})$ is again a Hopf algebra.

Given H, H' fin. dim. Hopf algebras, $\mathrm{rep}_H \simeq \mathrm{rep}_{H'}$ as tensor categories $\iff \exists F : H' = H_F$.

Note: There are tensor categories with no fiber functors.

Braided monoidal categories (categorical versions of *abelian* groups).

A monoidal category $\mathcal{C}$ is *braided* if has a *commutativity* constraint (the braiding), that is a nat. isom. $c_{X,Y} : X \otimes Y \to Y \otimes X$ s. t. the hexagon axioms hold, i. e. (3), (4) are commutative:

$$\begin{tikzcd}
(X\otimes Y)\otimes Z \arrow[r, "c_{X,Y}\otimes \mathrm{id}_Z"] \arrow[d, "a_{X,Y,Z}"] & (Y\otimes X)\otimes Z \arrow[r, "a_{Y,X,Z}"] & Y\otimes(X\otimes Z) \arrow[d, "\mathrm{id}_Y\otimes c_{X,Z}"] \\
X\otimes(Y\otimes Z) \arrow[r, "c_{X,Y\otimes Z}"] & (Y\otimes Z)\otimes X \arrow[r, "a_{Y,Z,X}"] & Y\otimes(Z\otimes X)
\end{tikzcd} \tag{3}$$

$$\begin{tikzcd}
X\otimes(Y\otimes Z) \arrow[r, "\mathrm{id}_X\otimes c_{Y,Z}"] \arrow[d, "a^{-1}_{X,Y,Z}"] & X\otimes(Z\otimes Y) \arrow[r, "a^{-1}_{X,Z,Y}"] & (X\otimes Z)\otimes Y \arrow[d, "c_{X,Z}\otimes \mathrm{id}_Y"] \\
(X\otimes Y)\otimes Z \arrow[r, "c_{X\otimes Y,Z}"] & Z\otimes(X\otimes Y) \arrow[r, "a^{-1}_{Z,X,Y}"] & (Z\otimes X)\otimes Y
\end{tikzcd} \tag{4}$$

A *symmetric* category is a braided monoidal category s. t. $c_{Y,X}c_{X,Y} = \mathrm{id}_{X,Y}, \forall X, Y$; the braiding is then called a symmetry.

Examples:

- $\mathrm{Vec}_{\Bbbk}$ is a symmetric tensor category: $c_{X,Y} = \tau_{X,Y} : x \otimes y \mapsto y \otimes x$.

- G group, $\mathfrak{g}$ Lie algebra $\rightsquigarrow \mathrm{rep}_G, \mathrm{rep}_{\mathfrak{g}}$ are symmetric tensor categories with symmetry τ.

- More generally, if H is a cocommutative Hopf algebra, then rep_H is a symmetric tensor category with symmetry τ.

- G group.

Vec^G = category of G-graded v. sp., i. e. $V = \oplus_{g\in G} V_g$.

If $V, W \in \mathsf{Vec}^G \implies V \otimes W \in \mathsf{Vec}^G$, $(V \otimes W)_g = \oplus_{h\in G} V_g \otimes W_{h^{-1}g}$.

Assume G abelian. If $\chi : G \times G \to \Bbbk^\times$ satisfies $\forall g, k, \ell \in G$

$$\chi(g, k\ell) = \chi(g, k)\chi(g, \ell), \quad \chi(gk, \ell) = \chi(g, \ell)\chi(k, \ell),$$

then $c^\chi : V \otimes W \to W \otimes V$,

$$c^\chi(v_g \otimes w_k) = \chi(g, k) w_k \otimes v_g, \quad v \in V_g, w \in W_k,$$

is a braiding.

Remarks:

- c^χ is a symmetry iff $\chi(g,k) = \chi(k,g)\ \forall g,k \in G$.

- Any braiding in Vec^G is of the form c^χ for a unique χ.

- $G = \mathbb{Z}/2 = \{1,-1\}$, $\chi(-1,-1) = -1$, then $(\mathrm{Vec}^{\mathbb{Z}/2}, \chi)$ is the symmetric category of super vector spaces.

Quasi-triangular Hopf algebras (when rep_H has a braiding?)

H Hopf algebra.
A universal R-matrix for H is $\mathcal{R} \in (H \otimes H)^\times$ satisfying
(QT1) $(\Delta \otimes \mathrm{id})(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{23}$.
(QT2) $(\mathrm{id} \otimes \Delta)(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{12}$.
(QT3) $\tau\Delta(h) = \mathcal{R}\Delta(h)\mathcal{R}^{-1}$, $\forall h \in H$.

$\rightsquigarrow$ rep_H is a braided tensor category, with braiding

$$\begin{tikzcd}
V \otimes W \arrow[rr, "c_{V,W}"] \arrow[dr, "\text{action of} R"'] & & W \otimes V \\
& V \otimes W \arrow[ur, "\tau"'] &
\end{tikzcd}$$

(H, R) is a quasi-triangular Hopf algebra.

Examples:

• (Drinfeld) Any fin. dim Hopf algebra gives rise to a quasi-triangular Hopf algebra:

$H \rightsquigarrow D(H) = H^* \otimes H$ (Drinfeld double of H)

• $U_q(\mathfrak{g})$ is a quasi-triangular Hopf algebra

Justification: A braided vector space is a pair (V, c), where V is a v. sp. and $c : V \otimes V \to V \otimes V$ is a linear isom. such that

$$(c \otimes \mathrm{id})(\mathrm{id} \otimes c)(c \otimes \mathrm{id}) = (\mathrm{id} \otimes c)(c \otimes \mathrm{id})(\mathrm{id} \otimes c). \qquad \text{(braid equation)}$$

This is closely related to the quantum Yang-Baxter equation.

(V, c) braided v. s. $\rightsquigarrow$ reps. of the braid gp. $\mathbb{B}_N \to \mathrm{GL}(V^{\otimes n}), n \geq 2$.

(H, R) quasi-triang. Hopf alg., $V \in \mathrm{rep}_H \rightsquigarrow (V, c_{V,V})$ braided v. sp.

Any Hopf algebra gives a machine of solutions of the braid eqn.:

$$H \rightsquigarrow D(H) \rightsquigarrow V \in \mathrm{rep}_H, c_{V,V}.$$

The solutions associated to $U_q(\mathfrak{g})$ are very important in low dimensional topology and some areas of theoretical physics.

Short history:

• Symmetric tensor categories were first considered by Mac Lane and Benabou (1963).

• Braided tensor categories were first considered by Joyal and Street (1986-1993).

• Quasi-triangular Hopf algebras and many other contributions are due to Drinfeld, cf. his address at ICM 86, Berkeley.

<u>Ribbon categories</u> (when ${}^*V \simeq V^*$ in a braided category? Note ${}^*V \simeq V^* \iff V \simeq V^{**}$).

A *ribbon category* is a rigid braided tensor category $\mathcal{C}$ provided with a natural isom. $\delta_V : V \to V^{**}$ s. t.

$$\delta_{V\otimes W} = \delta_V \otimes \delta_W, \tag{5}$$

$$\delta_{\mathbf{1}} = \mathrm{id}, \tag{6}$$

$$\delta_{V^*} = (\delta_V^*)^{-1} \tag{7}$$

Note: In any rigid braided tensor category, there is a natural isom. $\psi_V : V^{**} \to V$ given by

$$\begin{tikzcd} V^{**} \arrow[r, "l_{V^{**}}"] \arrow[rrrrr, bend right, "\psi_V"'] & \mathbf{1}\otimes V^{**} \arrow[r, "b_V\otimes \mathrm{id}_{V^{**}}"] & V\otimes V^*\otimes V^{**} \arrow[r, "\mathrm{id}\otimes c^{-1}_{V^*,V^{**}}"] & V\otimes V^{**}\otimes V^* \arrow[r, "\mathrm{id}_V\otimes e_{V^*}"] & V\otimes \mathbf{1} \arrow[r, "r_V"] & V \end{tikzcd};$$

it satisfies $\qquad \psi_{V\otimes W} = c_{W,V}c_{V,W}(\psi_V \otimes \psi_W).$

Alternative definition: A *ribbon category* is a rigid braided tensor cat. $\mathcal{C}$ with a nat. isom. $\theta_V : V \to V$ s. t.

$$\theta_{V\otimes W} = c_{W,V}c_{V,W}(\theta_V \otimes \theta_W), \tag{8}$$
$$\theta_{\mathbf{1}} = \mathrm{id}, \tag{9}$$
$$\theta_{V^*} = \theta_V^*. \tag{10}$$

The equivalence is given by $\theta_V = \psi_V \delta_V$.

- θ_V is called the *twist*; (8), (9), (10) are the *balancing axioms*.

- Symmetric $\implies$ ribbon, with $\theta_V = \mathrm{id}_V$.

- The twist is not unique (easy example in $\mathrm{rep}_{\mathfrak{sl}(2)}$).

- Ribbon categories $\rightsquigarrow$ invariants of ribbon tangles (up to isotopy) (Reshetikhin-Turaev).

Let $\mathcal{C}$ be a ribbon category, $V \in \operatorname{Obj}\mathcal{C}$, $f \in \operatorname{End} V$. Define the trace of f as

$$\begin{tikzcd} \mathbf{1} \arrow[r, "b_V"] \arrow[rrr, bend right, "\operatorname{tr}_q f \in \operatorname{End}\mathbf{1} \simeq \Bbbk"'] & V \otimes V^* \arrow[r, "f \otimes \operatorname{id}"] & V \otimes V^* \arrow[r, "\delta_V \otimes \operatorname{id}_{V^{**}}"] & V^{**} \otimes V^* \arrow[r, "e_{V^*}"] & \mathbf{1}. \end{tikzcd}$$

Also, $\dim_q V = \operatorname{tr}_q \operatorname{id}_V$. Then $\dim_q(V \otimes W) = \dim_q V \dim_q V$.

Example: $\mathcal{C} = \operatorname{rep}_{U_q(\mathfrak{g})}$ is ribbon, where q is not a root of 1; $\delta_V = q^{2\rho}$, where ρ is the semi-sum of positive roots. Here

$$\operatorname{tr}_q(f) = \operatorname{tr}_V q^{2\rho} f, \qquad \dim_q v = \operatorname{tr}_V q^{2\rho}.$$

$\operatorname{rep}_{\mathfrak{u}_q(\mathfrak{g})}$ is ribbon, if q is a root of 1 (of odd order).

Let $\mathcal{C}$ be a finite ribbon category. Consider the subcategory $\mathcal{C}_0$ of objects V with $\dim_q V = 0$.

Localizing $\mathcal{C}$ at $\mathcal{C}_0$ we get a <u>new</u> tensor category.

Example: $\mathcal{C} = \mathrm{rep}_{\mathfrak{u}_q(\mathfrak{sl}(2))} \rightsquigarrow \mathcal{C}_q(\mathfrak{sl}(2))$ is a fusion ribbon category but **not** rep_H for any Hopf algebra.

If $(H, \mathcal{R})$ is a quasi-triangular Hopf algebra, when rep_H is ribbon?

Set $\mathcal{R} = \sum_i \mathcal{R}_i \otimes \mathcal{R}^i$ and $\mathcal{R}_{21} = \sum_i \mathcal{R}^i \otimes \mathcal{R}_i$. The *Drinfeld element* is $\mathbf{u} = \sum_i \mathcal{S}(\mathcal{R}^i)\mathcal{R}_i$. A ribbon element for $(H, \mathcal{R})$ is a <u>central</u> $\mathbf{v} \in H$ such that

$$\Delta(\mathbf{v}) = (\mathcal{R}_{21}\mathcal{R})^{-1}(\mathbf{v} \otimes \mathbf{v}), \tag{11}$$

$$\varepsilon(\mathbf{v}) = 1, \tag{12}$$

$$\mathcal{S}(\mathbf{v}) = \mathbf{v}, \tag{13}$$

$$\mathbf{v}^2 = \mathcal{S}(\mathbf{u})\mathbf{u}. \tag{14}$$

Define $\theta_V =$ action of $\mathbf{v}$ on V. Then rep_H is ribbon.

- Kauffman-Radford: criterium for a Drinfeld double $D(H)$ in terms of modular group-likes.
- Ribbon fusion categories with extra property (related to the fusion rules) $\rightsquigarrow$ modular tensor categories.

- *References:*

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III. Classification of Hopf algebras: state of the art. Goal: classify finite-dimensional Hopf algebras.

From now on $\Bbbk = \mathbb{C}$, or any algebraically closed field $\operatorname{char} \Bbbk = 0$.

Let C be a coalgebra, $D, E \subset C$. Then

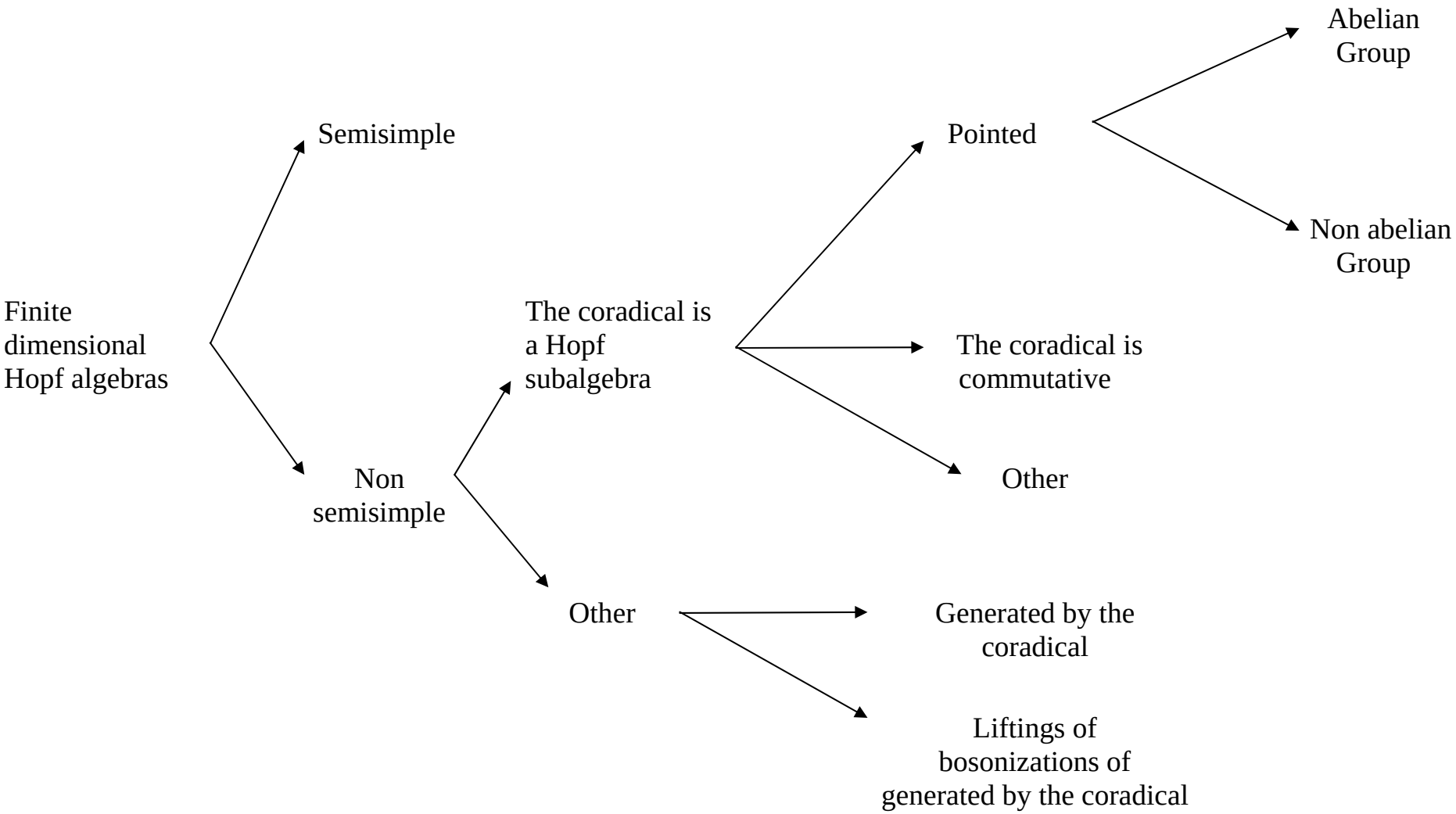
$D \wedge E = \{x \in C : \Delta(x) \in D \otimes C + C \otimes E\}$.

$\wedge^0 D = D$, $\wedge^{n+1} D = (\wedge^n D) \wedge D$.

Invariants of a Hopf algebra H:

- The coradical $H_0 =$ sum of all simple subcoalgebras of H.
- The *coradical filtration* is $H_n = \wedge^{n+1} H_0$.

Idea: approach the classification by the coradical.


Finite dimensional Hopf algebras
Semisimple
Non semisimple
The coradical is a Hopf subalgebra
Other
Pointed
The coradical is commutative
Other
Abelian Group
Non abelian Group
Generated by the coradical
Liftings of bosonizations of generated by the coradical

More invariants of a Hopf algebra H:

- The *Hopf coradical* $H_{[0]}$ is the subalgebra generated by H_0.

- The *standard filtration* is $H_{[n]} = \wedge^{n+1} H_{[0]}$.

- The *associated graded Hopf algebra* $\operatorname{gr} H = \oplus_{n\in\mathbb{N}} H_{[n]}/H_{[n-1]}$.

It turns out that $\operatorname{gr} H \simeq R\#H_{[0]}$, where

- $R = \oplus_{n\in\mathbb{N}} R^n$ is a graded connected algebra and it is a braided Hopf algebra.

Example:

$H = U_q(\mathfrak{b}) = \Bbbk\langle E, K, K^{-1} | KK^{-1} = 1 = K^{-1}K,\ KE = q^2 EK\rangle,$

$\Delta(K) = K \otimes K,\ \Delta(E) = E \otimes 1 + K \otimes E.$

- $H_0 = \Bbbk\langle K, K^{-1}\rangle = H_{[0]} \simeq \Bbbk\mathbb{Z}.$
- $H_n = H_{[n]} =$ subspace spanned by $K^j E^m,\ j \in \mathbb{Z}, m \leq n.$
- $H \simeq \operatorname{gr} H \simeq R\#\Bbbk\langle K, K^{-1}\rangle$, where
- $R = \Bbbk\langle E\rangle,\ c(E^i \otimes E^j) = q^{2ij} E^j \otimes E^i;\ \Delta(E) = E \otimes 1 + 1 \otimes E.$

Example: $H = U_q(\mathfrak{sl}(2))$

- $H_0 = \Bbbk\langle K, K^{-1}\rangle = H_{[0]} \simeq \Bbbk\mathbb{Z}$.

- $H_n = H_{[n]} =$ subspace spanned by $K^j E^i F^{m-i}$, $j \in \mathbb{Z}$, $i \in \mathbb{N}$, $m \leq n$.

- $\operatorname{gr} H = \Bbbk\langle X, Y, K, K^{-1} | KK^{-1} = 1 = K^{-1}K,$
$KX = q^2 XK,\ KY = q^{-2} YK,\ XY - qYX = 0\rangle$.

$$\Delta(X) = X \otimes 1 + K \otimes X,\ \Delta(Y) = Y \otimes 1 + K^{-1} \otimes Y.$$

- $R = \Bbbk\langle E\rangle$, $c(X \otimes Y) = q^2 Y \otimes X$, $c(Y \otimes X) = q^{-2} X \otimes Y$.

$\Delta(X) = X \otimes 1 + 1 \otimes X$, $\Delta(Y) = Y \otimes 1 + 1 \otimes Y$.

The following result generalizes previous work joint with H.-J. Schneider.

Theorem. (A.– Cuadra).

Any Hopf algebra with injective antipode is a <u>deformation</u> of the bosonization of <u>connected graded braided Hopf algebra</u> by a <u>Hopf algebra generated by a cosemisimple coalgebra</u>.

To provide significance to this result, we should address some fundamental questions.

Question I. Let C be a finite-dimensional cosemisimple coalgebra and $T : C \to C$ a bijective morphism of coalgebras. Classify all finite-dimensional Hopf algebras L generated by C such that $\mathcal{S}_{|C} = T$.

Question II. Given L as in the previous item, classify all finite-dimensional connected graded Hopf algebras in ${}^L_L\mathcal{YD}$.

Question III. Given L and R as in the previous items, classify all deformations, or liftings, H, that is, such that $\operatorname{gr} H \simeq R\#L$.

About Question I:

Theorem. (Stefan).

Let H be a Hopf algebra and C an $\mathcal{S}$-invariant 4-dim. simple subcoalgebra. If $1 < \operatorname{ord} \mathcal{S}^2_{|C} = n < \infty$, then there are a root of unity ω and a Hopf algebra morphism $\mathcal{O}_{\sqrt{-\omega}}(SL_2(\Bbbk)) \to H$.

- Classification of finite-dimensional quotients of $\mathcal{O}_q(SL_N(\Bbbk))$: E. Müller.

- Classification of quotients of $\mathcal{O}_q(G)$: A.–G. A. García.

About Question II:

The most important examples of connected graded braided Hopf algebras are Nichols algebras:

given a braided vector space (V, c), its Nichols algebra is a braided Hopf algebra

$$\mathfrak{B}(V) = \oplus_{n \in \mathbb{N}_0} \mathfrak{B}^n(V),$$

- $\mathfrak{B}^0(V) = \Bbbk$, $\mathfrak{B}^1(V) = V$,
- $\mathrm{Prim}(\mathfrak{B}(V)) = V$,
- V generates $\mathfrak{B}(V)$ as an algebra.

The most important examples of Nichols algebras are those of diagonal type: Let $\theta \in \mathbb{N}$ and $(q_{ij}) \in (\Bbbk^\times)^{\theta\times\theta}$. Consider

$$\chi : \mathbb{Z}^\theta \times \mathbb{Z}^\theta :\to \Bbbk^\times, \qquad \chi(\alpha_i, \alpha_j) = q_{ij},$$

where $\alpha_1, \dots, \alpha_\theta$ is the canonical basis. We also fix

- $X = \{x_1, \dots, x_\theta\}$ a basis of a vector space V of dimension θ.
- $\mathbb{X}$ = set of words in X (a basis of the tensor algebra $T(V)$).
- $T(V) = \oplus_{\alpha\in\mathbb{Z}^\theta} T^\alpha(V)$ the grading defined by $\deg(x_i) = \alpha_i$.
- For $u, v \in \mathbb{X}$, $q_{uv} = \chi(\deg u, \deg v)$.

Define a multiplication in $T(V) \otimes T(V)$ by

$$(u \otimes v)(w \otimes z) = q_{vw}\, uw \otimes vz, \qquad u, v, w, z \in \mathbb{X}$$

and an algebra map $\Delta : T(V) \to T(V) \otimes T(V)$ by
$\Delta(v) = v \otimes 1 + 1 \otimes v$, $v \in V$.
Proposition. (Lusztig, A.-Schneider). Given $a_1, \dots, a_\theta \in \Bbbk^\times$, there exists a unique bilinear form on $T(V)$ such that

$(x|yy') = (\Delta(x)|y \otimes y')$,

$(xx'|y) = (x \otimes x'|\Delta^{\mathrm{op}}(y))$,

$(x_i|x_j) = \delta_{ij} a_i$,

$(T^\alpha(V)|T^\beta(V)) = 0$,
for all $x, x', y, y' \in T(V)$, $1 \le i, j \le \theta$, $\alpha, \beta \in \mathbb{Z}^\theta$.

Definition. Let $\mathcal{J} = \mathcal{J}(V) =$ radical of $(\,|\,)$, a 2-sided ideal of $T(V)$. The quotient $\mathfrak{B}(V) := T(V)/\mathcal{J}$ is the **Nichols algebra** associated to the matrix (q_{ij}).

Problem I. Classify all matrices $(q_{ij})_{1\leq i,j\leq\theta}$ such that $\dim \mathfrak{B}(V) < \infty$.

Problem II. For such Nichols algebras, give a nice set of generators of the ideal $\mathcal{J}$ (defining relations).

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Problem I was solved in

I. Heckenberger, *Classification of arithmetic root systems*, Adv. Math. **220** (2009) 59–124.

Problem II was solved in

I. Angiono. *A presentation by generators and relations of Nichols algebras of diagonal type and convex orders on root systems.* `arXiv 1008.4144v1`. 29 pages. Submitted.

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